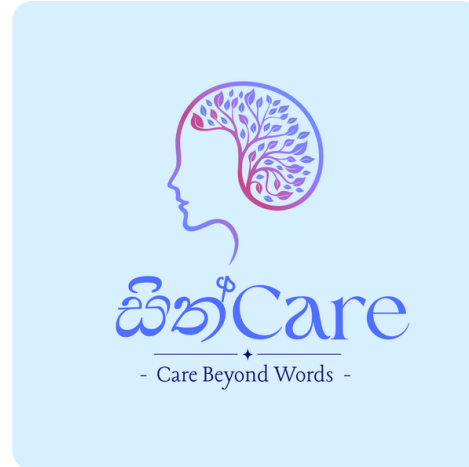




Group 10

SithCare

Mental Health Channeling Platform

Care Beyond Words

SENG 31242 – System Design Project

Bachelor of Science (Hons.) in Software Engineering

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Supervisor: Eng. Dr. Tiroshan Madushanka

Date: 21 June 2026

Presentation Roadmap

01 Problem & Solution

Mental health access gap in Sri Lanka and how SithCare addresses it

02 System Overview

User roles, core features and system scope

03 Architecture

Layered MVC design, MERN stack and system components

04 Key Workflows

Appointment booking, telemedicine and emergency support flows

05 Security & Design

Security architecture , non-functional design decisions , UI wireframes

06 Timeline & Future

Project schedule and planned enhancements

The Problem Context

Current Challenges in Sri Lanka

Severe Shortage

- Only **2.5 mental health workers** per 100,000 population (WHO 2024)

Access Barriers

- No centralized directory
- Phone booking stigma
- Language barriers

Poor Patient Experience

- No digital queue tracking
- no telemedicine for rural areas
- 1-4 hour waits

Crisis Support Gap

- Crisis helplines unknown to most patients
- no digital emergency access

Our Research Foundation

150+

Survey respondents via Google Forms

5+

Interviews conducted for Mental Patients

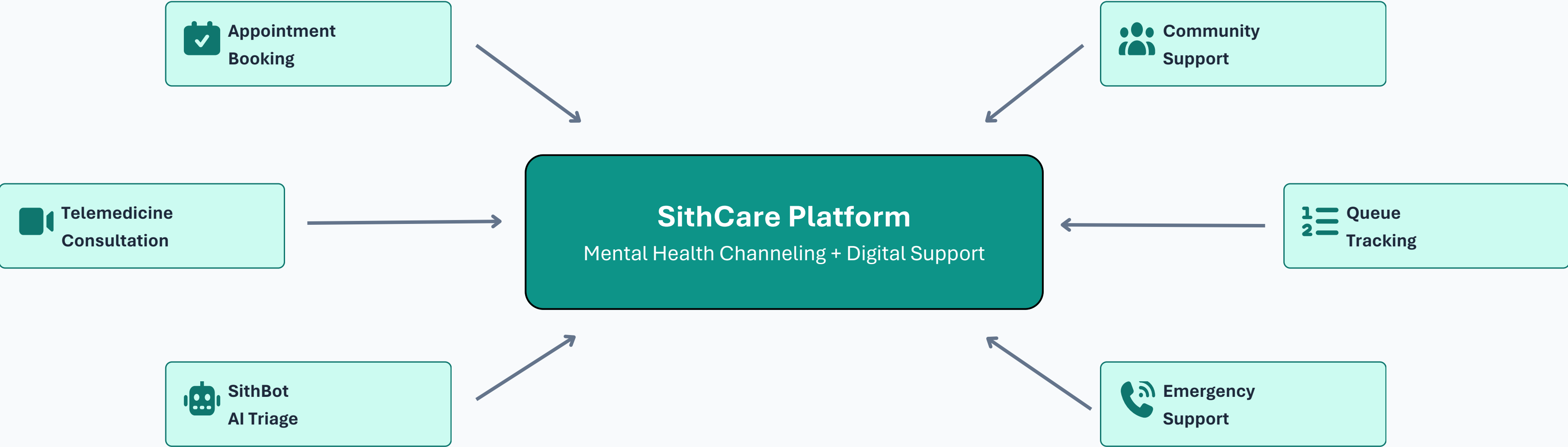
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Mental health specialist consultations

4

Competing platforms analyzed

SithCare - Our Solution



Key Differentiators

Full Sri Lankan localization (Sinhala, Tamil, English) | LKR currency via PayHere | Multi-clinic management | Adult Mode for elderly users | Anonymous community | Volunteer listener support |



Scope & User Roles



Patient

- Search specialists
- Book appointments
- Pay
- Join consultations
- Track queue



Specialist

- Manage availability
- Conduct consultations
- View patient records



Clinic Admin

- Manage doctors
- Appointments
- Queues
- Payments
- Clinic operations



Platform Admin

- Onboard clinics
- Manage users
- System analytics
- Compliance oversight



Moderator (Future Update)

- Review community posts
- Manage harmful content
- Ensure safety

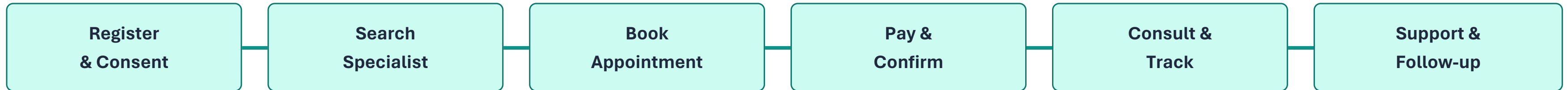


Volunteer Listener (Future Update)

- Join listener sessions
- Access training
- Provide peer support

All roles use **Role-Based Access Control (RBAC)** with JWT authentication. Each role has a dedicated dashboard with functions specific to their responsibilities.

Core System Features



SithBot AI Triage

AI-powered chatbot for specialist recommendations and mental health guidance

Emergency Support

24/7 crisis helplines with consent-based location sharing and urgent slots

Anonymous Community

Safe space for mental health discussions with complete anonymity

Multilingual Access

Full support for Sinhala, Tamil and English with Adult Mode for elderly users

SMS/Email Alerts

Automated reminders for appointments, queue updates and payments via Twilio

Digital Receipts

Automated PDF receipt generation for all payments via PayHere gateway

52 functional requirements documented across 11 categories including authentication, appointment management, telemedicine, payments, emergency support, AI chatbot and community features.



High-Level Architecture



Presentation Layer

React SPA with Tailwind CSS - responsive, role-based dashboards



Application / Controller Layer

Express.js REST APIs with JWT authentication and RBAC middleware



Business Logic Layer

Node.js services - appointments, queue, payments, AI triage, notifications



Data Access Layer

MongoDB Atlas with Mongoose ODM - cloud-native, flexible schema

Quality Attributes

Scalability

500 concurrent users, stateless APIs

Security

JWT + RBAC + TLS 1.3 + AES-256

Maintainability

Modular MVC, single language (JS)

Reliability

99.5% uptime target, service separation

Usability

Sub-3s page loads, Adult Mode, responsive

Design Decisions :

- Layered MVC architecture with MERN stack enables parallel development
- Future migration path toward microservices.

- Clear separation of concerns

Technology Stack



01

Frontend

React + Vite

Tailwind CSS

Responsive SPA

Role-based UI

02

Backend

Node.js

Express.js

REST API

JWT Sessions

03

Database

MongoDB Atlas

Mongoose ODM

Cloud Storage

Flexible Schema

04

Security

JWT + bcrypt

TLS 1.3

RBAC

OWASP Compliant

05

Services

PayHere (Payments)

Jitsi Meet (Video)

Twilio (SMS)

Google Maps

Cloud Deployment:

- Frontend on Vercel/Netlify (CDN)
- Backend on Render/Railway
- MongoDB Atlas Free Tier
- GitHub Actions CI/CD pipeline



System Context & Components

Frontend Components

- Auth Module (registration, login, OTP)
- Patient Dashboard (booking, tracking)
- Clinic Admin Panel (operations)
- SithCare Admin Panel (analytics)
- Doctor Portal (schedules, consultations)
- SithBot Widget (AI triage)

Backend Services

- Auth Controller (JWT, RBAC)
- Appointment Service (booking)
- Queue Manager (live tracking)
- Notification Service (SMS/email)
- Payment Gateway Adapter (PayHere)
- Emergency Handler (crisis response)

External Integrations

PayHere

Jitsi Meet

Twilio SMS

Nodemailer

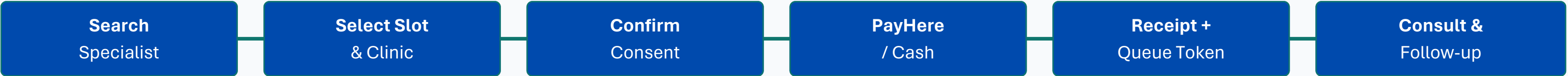
Google Maps

MongoDB Atlas

Key Workflows



Workflow 1: Appointment Booking



Workflow 2: Telemedicine Consultation

Join Link → Jitsi Room → Consultation Record

Phone fallback available for low bandwidth situations

Workflow 3: Emergency Support

Risk Signal → Consent Prompt → Helpline / Urgent Slot

Location sharing only with explicit user confirmation

Complete Use Case Coverage

- **UC-01** Register & Login
- **UC-02** Profile & Consent
- **UC-03** Search Specialist
- **UC-04** Book Appointment
- **UC-05** Manage Availability
- **UC-06** Approve/Reschedule
- **UC-07** Telemedicine
- **UC-08** Payment & Receipts
- **UC-09** Live Queue Tracker
- **UC-10** Emergency Support
- **UC-11** Anonymous Community
- **UC-12** SithBot AI Triage



Security Design



Authentication

JWT sessions with 15-min expiry, bcrypt password hashing, OTP verification, account lockout after 5 failed attempts



Authorization

RBAC for 6 distinct roles with dedicated dashboards and route guards.
Role-based middleware enforces permissions



Data Protection

TLS 1.3 for data in transit, AES-256 for sensitive records at rest, no card data stored (PayHere handles PCI-DSS)



OWASP Mitigation

All OWASP Top 10 risks addressed: injection, broken auth, sensitive data exposure, XXE, broken access control



SithBot Safety Rules

- Triage guidance only - no clinical diagnosis
- Crisis keyword detection routes to emergency support.
- Mandatory medical disclaimers on every response.

Non-Functional Design Decisions

Performance

React SPA with CDN hosting. Sub-3 second page loads. Efficient REST APIs with async processing.

Reliability

Service separation with fallback handling. 99.5% uptime target. Graceful degradation on third-party failures.

Scalability

Stateful REST APIs. MongoDB Atlas cloud database. Supports 500 concurrent users with horizontal scaling.

Security

RBAC, JWT tokens, TLS encryption, input validation, comprehensive security logging and monitoring.

Usability

Simple workflows, responsive UI, Adult Mode with 18px font for elderly users, low cognitive load design.

Maintainability

Layered MVC architecture with modular service packages. 70% unit test coverage target. GitHub CI/CD.

- **20 Non-Functional Requirements** documented across 6 categories.
- Every design decision is traceable to specific NFRs through an explicit mapping framework defined in the SDS.

User Interface Design



1. SithCare Admin

<https://www.figma.com/design/KVuluJcvd5DSvpkDFwSkxb/Admin-Dashboard-Wireframe?node-id=0-1&t=VXzz72Bfdqolyjne-1>

2. SithCare Clinic

<https://www.figma.com/files/team/1645559070070455823/project/611206225?fuid=1493315410793033636>

3. SithCare Patients

<https://www.figma.com/design/gAGRnLYlmWepWosPffJ7H4/Untitled?node-id=0-1&t=g6XoQPqgiSUrAmYY-1>



Project Timeline & Future Work

Development Timeline



Future Enhancements

Native Mobile Apps

Android and iOS applications for broader accessibility

EHR Integration

Link with government healthcare record systems

Advanced Analytics

Predictive insights on service demand and risk trends

Wearable Support

Optional sensor-based wellbeing signal integration

Expanded AI Support

NLP-powered SithBot with sentiment analysis and safer guidance

Mood Tracking

Patient progress monitoring for longitudinal mental health care



Thank You!

Q & A

SithCare - Care Beyond Words

Group 10 | University of Kelaniya

SENG 31242 - System Design Project

Supervisor: Eng. Dr. Tiroshan Madushanka